

BatsiraiMalcolm Dzimati

Software Engineer | Distributed Systems & Cloud-Native Backend | Java, Python, .NET, TypeScript

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Summary

Software Engineer with 3+ years of experience building scalable, high-performance distributed systems across financial services, cloud-native platforms, and data-intensive environments. Proven track record delivering production-grade backend systems, event-driven architectures, and high-volume transactional platforms in critical, time-sensitive environments. Experienced across the full software development lifecycle with hands-on expertise in Java, Python, .NET, cloud infrastructure (AWS & Azure), DevOps, and modern engineering practices. Proven ability to lead technical initiatives, architect scalable solutions, and translate complex operational challenges into reliable, high-availability systems.

Skills

Languages	Java, Python, C#, TypeScript
Frameworks	Spring Boot, FastAPI, Django, .NET, Angular, Next.js
Cloud	AWS (Solutions Architect), Microsoft Azure (DevOps Engineer Expert, Developer Associate)
Architecture	Distributed Systems, Event-Driven Architecture, Microservices, API Design, Scalable System Design, Event Streaming
MLOps, DevOps & Infra	CI/CD, Kubernetes, Helm, Infrastructure as Code (Bicep), Observability
Testing	TDD, Unit Testing, Integration Testing, K6 (load testing), E2E Testing (SpecFlow, xUnit, Playwright)
AI	LLM Applications, Retrieval-Augmented Generation (RAG), Agentic Workflows, LangGraph, MCP Tooling
Data	PostgreSQL, SQL, Snowflake, Query Optimisation, Schema Design
Leadership & Delivery	Technical Leadership, Cross-Functional Collaboration, Stakeholder Communication, Agile Delivery, Technical Mentorship

Experience

Vectr Labs

Sandton, Gauteng

FORWARD DEPLOYED TECH LEAD – AI & CLOUD PLATFORMS

October 2025 – Present

- Joined as one of the earliest technical hires; led architecture, cloud infrastructure strategy, AI initiatives, and full-stack engineering delivery across multiple client and internal platform projects
- Engaged directly with client engineers and stakeholders to understand existing AWS and Azure environments, architectural standards, and technology roadmaps, designing bespoke applications and integrations that aligned with platform constraints and long-term cloud strategy.
- Architected and delivered an intelligent analytics dashboard for a major South African insurance company, processing millions of records to surface key business insights through AI-powered anomaly detection, trend analysis, and automated reporting — built on distributed backend services, Snowflake, and cloud-native APIs
- Architected near real-time telemetry and predictive maintenance platforms processing operational data from 160+ industrial assets across 10+ integrated systems using event-driven architecture, Kafka, Snowflake, and cloud-native backend services (Python/FastAPI)
- Built AI-powered RAG and agentic workflows (LangGraph, MCP tooling, vector retrieval) to generate maintenance diagnostics, repair recommendations, and automated fleet health summaries from OEM documentation and live operational telemetry
- Established organisation-level DevOps and CI/CD standards: Infrastructure as Code, automated testing pipelines, observability practices, and cloud deployment processes across AWS and Azure environments
- Mentored 10+ junior engineers through code reviews, architectural guidance, and engineering development planning

Investec Private Bank

Sandton, Gauteng

INTERMEDIATE SOFTWARE ENGINEER – PAYMENTS & CRYPTOGRAPHY PLATFORM

April 2024 – December 2025

- Core contributor to high-stakes payments infrastructure, operating on a formal on-call support rotation for critical card, PIN, and cryptography systems — responding rapidly to live production incidents in a zero-tolerance, high-pressure financial environment
- Delivered payment system modernisation across Java (Spring Boot) and .NET microservices, improving scalability and transaction throughput during peak load events including Black Friday; collaborated with architects and tech lead to evaluate cloud-native alternatives for legacy payment components
- Designed and implemented event-driven architecture patterns (replacing synchronous polling flows) to improve system responsiveness and reduce load under peak transaction volumes — a pattern later adopted across multiple payment services
- Built and maintained cryptographic infrastructure integrating HSMs (PayShield, Luna) for secure PIN operations (generation, verification, change) and key lifecycle management, ensuring compliance with PCI-DSS and payment security standards
- Developed backend microservices using Java (Spring Boot), C# (.NET), SQL, and Azure supporting high-volume, low-latency financial transaction flows
- Applied rigorous TDD practices with comprehensive automated test coverage: unit, integration, and end-to-end tests using SpecFlow, xUnit, k6 (load testing), and Playwright
- Collaborated with internal security, vendor, and compliance teams to maintain reliability and security of card authentication and token management systems

Investec Private Bank

Sandton, Gauteng

GRADUATE SOFTWARE ENGINEER – PAYMENTS & CRYPTOGRAPHY

Jan 2023 – March 2024

- Planned and led cryptographic key rotation ceremonies for critical banking systems, coordinating cross-functional teams and HSM infrastructure changes to ensure secure key migration with zero customer-facing downtime.
- Built and maintained RESTful APIs supporting secure payment and card authentication systems
- Developed systems for secure PIN operations (generation, verification, change) and integrated with HSM infrastructure and end-to-end encryption pipelines
- Contributed to cloud migration initiatives and CI/CD pipeline implementation, modernising legacy payment components
- Supported credit card operations and 3D Secure authentication flows

Education

University of Pretoria

Pretoria, Gauteng

BScHONS IN COMPUTER SCIENCE

Jan 2023 – Dec 2024

University of Pretoria

Pretoria, Gauteng

BSc IN COMPUTER SCIENCE & STATISTICS

Jan 2020 – Dec 2022

Selected Projects

Intelligent Insurance Analytics Dashboard

PYTHON, SNOWFLAKE, CLOUD-NATIVE APIs, AI/ML

- Designed and delivered a distributed analytics platform for a major South African insurer, processing millions of records to surface key business insights
- Implemented AI-powered anomaly detection, automated trend analysis, and executive reporting dashboards on cloud-native back-end services

Cryptography Platform

C#, .NET, JAVA, HSM, SQL, AZURE

- Maintained and improved a high-performance cryptography platform underpinning client logins and SOS, the bank's keys to the clearing house, all PIN operations, and bulk card printing
- Ensured low-latency, highly reliable cryptographic services under heavy transactional load, integrating with HSMs (PayShield, Luna) for secure key management

Secure PIN Change Kiosk

C#, .NET, ANGULAR, HSM, SECURITY

- Built a full-stack PIN change kiosk application (Angular front end, C#/.NET back end) enabling clients to securely change and verify card PINs, integrating directly with HSM infrastructure for cryptographic PIN operations
- Designed the application with end-to-end encryption and a security-first architecture, ensuring sensitive PIN data was never exposed in transit or at rest and remained compliant with payment security standards

Predictive Maintenance Platform

PYTHON, FASTAPI, KAFKA, SNOWFLAKE, CLOUD

- Built event-driven backend services to process large-scale IoT telemetry data from 160+ industrial assets
- Enabled real-time trend analysis and root cause investigation across operational systems

3D Secure Optimisation

PAYMENTS, EVENT-DRIVEN ARCHITECTURE, JAVA

- Replaced synchronous polling with event-driven architecture, improving authentication responsiveness
- Reduced system load during peak transaction periods

Payment Modernisation

C#, JAVA, SQL, AZURE

- Improved transaction reliability and scalability across core payment systems
- Optimised processing flows, reducing end-to-end latency

Token Management Platform

PAYMENTS, TOKENISATION, HSM, SECURITY

- Built token issuance, rotation, and revocation workflows integrating with HSM cryptographic infrastructure
- Ensured compliance with payment security standards for secure handling of sensitive card data

Certifications

AWS Certified Solutions Architect – Associate , AWS Certified	<i>May 2026</i>
Azure DevOps Engineer Expert (AZ-400) , Microsoft Certified	<i>Aug 2025</i>
Azure Developer Associate (AZ-204) , Microsoft Certified	<i>Mar 2024</i>
Azure Fundamentals (AZ-900) , Microsoft Certified	<i>Jun 2023</i>
Cryptography & Secure Systems , Udemy	<i>2023</i>